

Elucidating the Intersystem Crossing of the Nitrogen-Vacancy Center up to Megabar Pressures

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The integration of nitrogen-vacancy color centers into diamond anvil cells has opened the door to quantum sensing at megabar pressures. Despite a multitude of experimental demonstrations and applications ranging from quantum materials to geophysics, a detailed microscopic understanding of how stress affects the NV center remains lacking. In this work, using a combination of first principles calculations as well as high-pressure NV experiments, we develop a complete description of the NV's optical properties under general stress conditions. In particular, our *ab initio* calculations reveal the complex behavior of the NV's intersystem crossing rates under stresses that both preserve and break the defect's symmetry. Crucially, our proposed framework immediately resolves a number of open questions in the field, including (i) the microscopic origin of the observed contrast enhancement in (111)-oriented anvils, and (ii) the surprising observation of NV contrast inversion in certain high-pressure regimes. Our work lays the foundation for optimizing the performance of NV high-pressure sensors by controlling the local stress environment, and more generally, suggests that symmetry-breaking stresses can be utilized as a novel tuning knob for generic solid-state spin defects.

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Pressure represents a powerful tuning knob for condensed matter systems, enabling access to novel physical states, ranging from record-high temperature superconductivity [1] to exotic structural phases [2]. Access to megabar pressures [3] in the laboratory is enabled by the diamond anvil cell (DAC) [Fig. 1(a)], an apparatus consisting of two opposing diamond tips that compress a small sample within a gasketed chamber. However, the DAC imposes severe constraints on metrology. Perhaps the most important is the inability to perform spatially-resolved local measurements of the physics inside the high-pressure chamber [4–6].

To this end, a tremendous amount of excitement has centered on the integration of nitrogen vacancy (NV) color centers into diamond anvil cells [12–20]. By directly implanting such spin-defect sensors into the anvil tip

(i.e., culet) applying the pressure [12–14], seminal recent experiments have demonstrated the ability to image local stresses and magnetism with submicron resolution [21], and resolving heterogeneities in functional materials is now becoming possible [22]. This approach has had an almost immediate impact on our understanding of multiple families of materials under pressure, ranging from hydride [17,23,24] and nickelate [22,25] superconductors to magnetic minerals [18].

Despite these successes, our microscopic understanding of the NV center under pressure remains relatively nascent, with two broad sets of open questions. First, it is generally believed that the stress environment must be carefully managed in order to enable high-pressure NV quantum sensing [Fig. 1(b)] [26–29]. As an example, the NV's optical contrast depends sensitively on its crystallographic orientation relative to the culet [17–19,30]. However, the underlying reasons for this sensitivity—and whether more optimal stress conditions exist—remain unclear. Second, a

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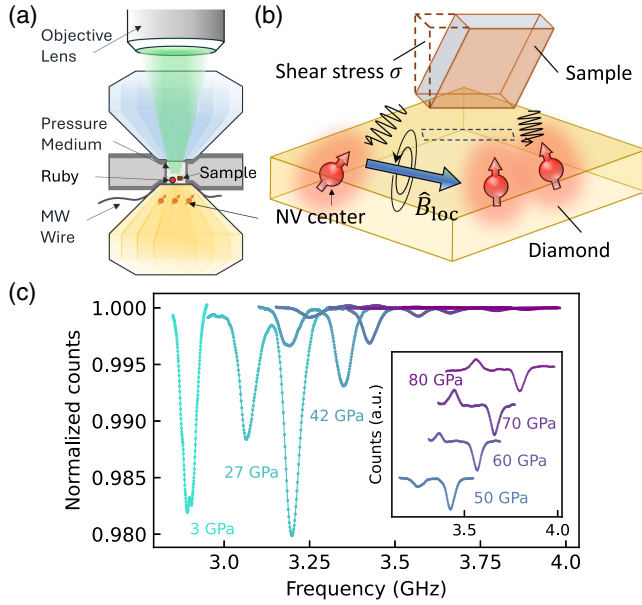


FIG. 1. (a) Schematic of the diamond anvil cell (DAC) geometry. The DAC sample chamber is defined by the gasket-anvil assembly; it is loaded with the sample of interest, pressure-transmitting medium, and a ruby microsphere. A ~ 50 -nm layer of NV centers (about 1 ppm density) is embedded into the diamond anvil directly below the sample chamber. For ODMR measurements, a platinum wire is placed on the bottom culet to deliver microwaves. (b) Major quantum sensing applications using the NV center include magnetometry [7] and sensing normal and shear (depicted) stresses σ in the sample [8–11]. (c) Continuous-wave ODMR measurements of NV centers in the (100)-cut anvil exhibit a drastic reduction in contrast with increasing pressure. The dominant culet stresses have symmetry-preserving and breaking projections on all NV subgroups, thereby inducing both a blue shift, Π_z , and a splitting, $2\Pi_\perp$ in the ODMR peaks. Notably, a surprising inversion of contrast is observed on the left peak around 60 GPa, as shown in the inset.

multitude of experiments across a variety of conditions [16,17], have all observed the puzzling *inversion* of NV contrast under pressure [Fig. 1(c)] [17]. On the one hand, this inversion complicates signal extraction from the NV center; on the other, it may offer metrological advantages of its own, such as extending the range of stresses over which NVs remain measurable, as demonstrated in Ref. [22]. Taken together, these questions point to the importance of developing a microscopic framework for understanding and predicting the properties of NV centers in a generic stress environment.

In this Letter, we combine an extensive set of first principles calculations with high-pressure NV experiments on three different culet orientations [i.e., (100)-, (110)-, and (111)-oriented anvils]. Our *ab initio* simulations allow us to estimate two crucial sets of NV parameters as a function of the stress tensor: (i) the rates of intersystem crossing (ISC), a nonradiative transition between electronic states with different spin multiplicities and (ii) the spin polarization in

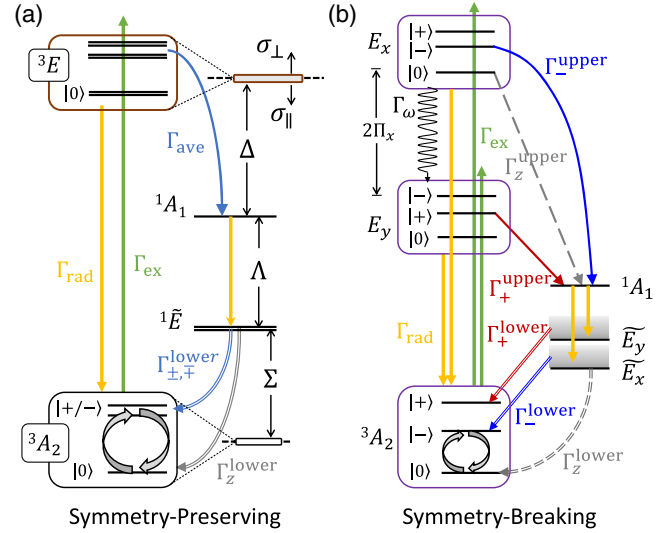


FIG. 2. The negatively charged NV center’s energy level diagram and its optical cycle under (a) symmetry-preserving and (b) symmetry-breaking stress (here Π_x stress, defined in Supplemental Material [31]). The NV center’s low-lying electronic states contain two spin triplets 3A_2 and 3E , and two spin singlets 1A_1 and 1E , with three energy gaps defined as Δ , Λ , Σ in (a). The spin-1 basis adopted here is $|m_s = 0\rangle$, and $|m_s = \pm\rangle = (1/\sqrt{2})(|m_s = 1\rangle \pm |m_s = -1\rangle)$, which are the spin eigenstates under Π_x stress. Notably, for symmetry-preserving stress, $\sigma_\perp = \frac{1}{2}(\sigma_{xx} + \sigma_{yy})$ and $\sigma_\parallel = \sigma_{zz}$ play qualitatively different roles in shifting energy gaps. Symmetry-breaking stress, however, breaks the defect’s point group symmetry and allows every spin state to participate in the optical cycle. The ISC rates are color coded for their spins, with blue, dark red, and gray for $|m_s = -, +, 0\rangle$, respectively in (b). The symmetry-allowed transitions are plotted and the line styles denote the microscopic origin for these ISCs [31].

the ground-state manifold. This enables us to propose and analyze a microscopic model that characterizes the NV’s optically detected magnetic resonance (ODMR) contrast under general stress conditions. Our main results are twofold. For stress environments which preserve the C_{3v} symmetry of the NV center, we predict that the optical contrast is mainly determined by the “upper” intersystem crossing rate, Γ_{ave} [Fig. 2(a)]. To test these predictions, we directly compare to DAC measurements exhibiting a range of different hydrostaticities [Fig. 3(b)] [15–18]. For symmetry-breaking stresses, we uncover a subtle interplay between the stress-induced spin-orbit coupling (SOC) and the Jahn-Teller (JT) effects of the NV center. This interplay causes a nonmonotonic response of the NV center’s “lower” ISC rate, Γ_z^{lower} [Fig. 2(b)] as a function of stress (Fig. 4), and ultimately produces an unconventional polarization mechanism that yields the observed contrast inversion. While our *ab initio* simulations focus on the (100)-oriented culet, our proposed mechanism for contrast inversion should also apply to both (110)- and (111)-oriented culets. To this end, we perform experiments

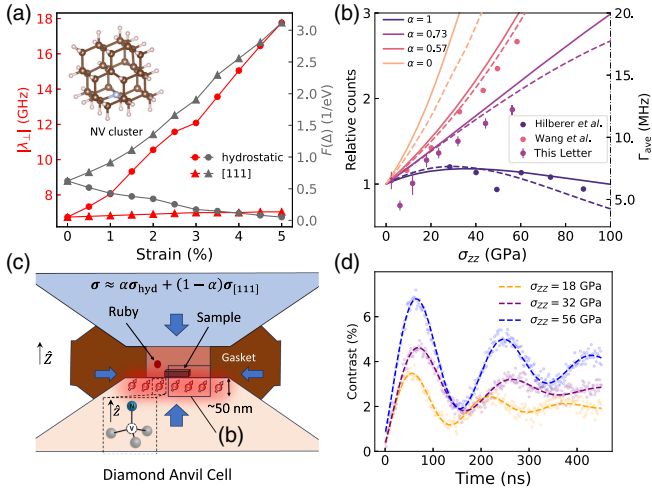


FIG. 3. (a) Computations of the intermediate components of the upper ISC, i.e., transverse spin-orbit coupling $\lambda_{\perp}$ (red) and vibrational overlap $F(\Delta)$ between the 3E and 1A_1 manifolds (gray) versus hydrostatic (circle) and uniaxial [111] strain (triangle). The inset shows a cluster model of the NV center, which we use as a basis for our *ab initio* calculations. (b) Upper ISC rate Γ_{ave} assembled from $\lambda_{\perp}$ and $F(\Delta)$ (dashed), and comparison of the contrast (relative to that at the ambient condition) between simulation (solid) and experiments [16,18] (dots), where the color codes the hydrostaticity α . Notably, the ISC rate exhibits a strong correlation with the relative contrast. (c) Schematic of a close-up DAC with (111)-cut diamond, and the embedded NV centers. (d) Rabi oscillations of the [111] NV at $\sigma_{ZZ} = 18, 32, 56$ GPa, respectively, from which contrast is extracted. The experimental data are fitted by damped sine waves and plotted by the orange, purple, and blue dashed lines.

on both of these anvil cell geometries and indeed observe the predicted contrast inversion (Fig. 5).

Microscopies of the NV's ODMR contrast—Each NV center hosts a spin-1 electronic ground state that can be optically polarized and read out [53,54]. Here, we will work in the spin triplet basis $|m_s = 0, +, -\rangle$, where $|m_s = \pm\rangle = (1/\sqrt{2})(|+1\rangle \pm |-1\rangle)$ and $|\pm 1\rangle$ are the familiar Zeeman eigenstates (where the quantization axis is defined along the NV axis).

NV center metrology is primarily performed via ODMR spectroscopy, where a 532-nm laser first excites the NV center and polarizes its population into the $|m_s = 0\rangle$ spin sublevel [53–55]. Microscopically, this polarization arises because the $|m_s = 0\rangle$ sublevels in 3E are forbidden to intersystem cross (into the 1A_1 manifold) at leading order [56,57], while the lower intersystem crossing rates (Γ^{lower} , Fig. 2) exhibit a weak spin dependence [58]. Thus, during each optical cycle, population is preferentially transferred into the $|m_s = 0\rangle$ ground state; this yields the conventional experimental observation of $\gtrsim 70\%$ spin polarization [59,60].

Crucially, the same optical pathway also naturally leads to spin-dependent fluorescence, enabling optical readout of

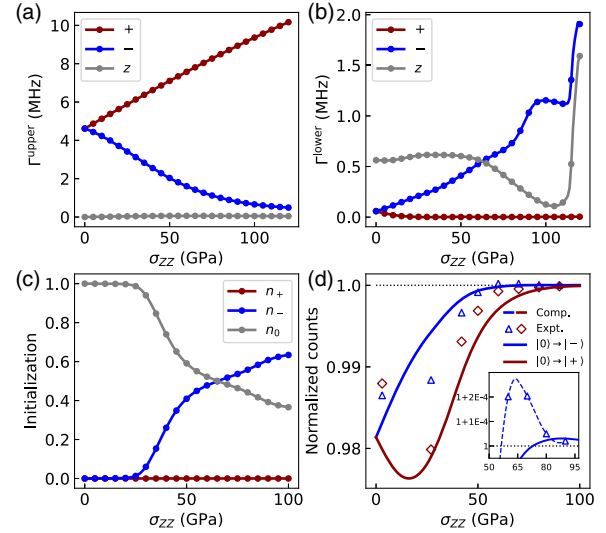


FIG. 4. Computations of the ISC rates and ODMR contrast of NV centers in the (100)-cut diamond under stress. (a) Upper ISC rates versus stress with color codes for the three spins. (b) Lower ISC rates versus stress, with Γ_z^{lower} exhibiting non-monotonic trend coming from negative interference between different ISC mechanisms (see main text and SM). (c) Ground state population distribution among the three spins. A gradual transfer from n_0 to n_- begins around 25 GPa and n_- dominates the population from 65 GPa. (d) ODMR contrast from simulations (curve) obtained by solving the rate model, and from experiments (dot), which is also shown in Fig. 1(c). Notably, the contrast inversion in the left peak (representing transitions $|m_s = 0\rangle \leftrightarrow |m_s = -\rangle$ driven by the MW) observed from experiments [17] (discrete) is reproduced, as shown in the inset. The “predicted” onset of positive contrast is slightly later compared to experiments, and the magnitude is also smaller, with possible reasons for this discrepancy discussed in detail in the Supplemental Material [31].

the NV's magnetic resonance spectra. In particular, since the $|m_s = 0\rangle$ sublevels in 3E exhibit an extremely small upper intersystem crossing rate, their dominant dynamics correspond to radiative relaxation directly to the ground state; since the laser excitation is spin preserving, this immediately implies that the $|m_s = 0\rangle$ ground state is “brighter” than the $|m_s = +, -\rangle$ states. ODMR spectroscopy proceeds by measuring the magnitude of the NV's fluorescence dip (i.e., contrast) when a microwave field resonant with the spin transition is applied (compared to when it is off). To this end, the NV center's contrast is controlled by two key ingredients, both determined by the ISC rates (Fig. 2): (i) the degree of spin polarization and (ii) the relative brightness of the three spin sublevels.

We evaluate the upper ISC rates using Fermi's golden rule [57], i.e., $\Gamma = (2\pi/\hbar)|\lambda|^2 F(\Delta)$, where the matrix element, $\lambda = \langle \psi_{\text{final}} | H_{\text{so}} | \psi_{\text{init}} \rangle$, arises from spin-orbit interactions and the vibrational overlap function $F(\Delta)$ characterizes the density of states at the gap Δ [Fig. 2(a)]. By comparison, the lower ISC rates are significantly more complicated, since they are forbidden at first order [31].

Optimizing NV contrast for symmetry-preserving stresses—We investigate a general symmetry-preserving stress of the form: $\sigma = \alpha\sigma_{\text{hyd}} + (1 - \alpha)\sigma_{[111]}$, where α characterizes the degree of hydrostaticity, and $\sigma_{\text{hyd}}, \sigma_{[111]}$ represent the hydrostatic and uniaxial [111] stresses, respectively. Since the symmetry of the NV center is preserved, the optical cycle is qualitatively the same as the ambient case.

To estimate the intersystem crossing rates, we compute the transverse SOC $\lambda_{\perp}$ and $F(\Delta)$ [Fig. 3(a)] as a function of strain (and then convert to stress), for both the uniaxial [111] and hydrostatic cases (i.e., $\alpha = 0, 1$ respectively). A few remarks are in order. First, $\lambda_{\perp}$ (red) increases with compression in both hydrostatic and uniaxial [111] environments, although the effect is significantly stronger for the former. Second, we find that $F(\Delta)$ (gray) exhibits opposite trends for the two types of stress environments, implying that the vibrational overlap increases significantly with uniaxial [111] strain, but is suppressed by hydrostatic strain [61].

Using $\lambda_{\perp}$ and $F(\Delta)$, we now compute the upper ISC rates, Γ_{ave} , versus stress [dashed curves, Fig. 3(b)]. For hydrostatic stress (dark purple, $\alpha = 1$), the ISC rate exhibits a non-trivial trend, with a peak value at approximately ~ 30 GPa. Interestingly, this is a manifestation of competition between the behaviors of $\lambda_{\perp}$ and $F(\Delta)$, where the former dominates at small stresses while the latter controls the large stress limit. For smaller α (i.e., a larger uniaxial component), Γ_{ave} exhibits a more monotonic behavior as a function of stress.

In order to predict how the NV's ODMR contrast changes versus stress, we directly solve the rate-equation model for the NV's optical cycle [Fig. 2(a)] utilizing these computed ISC rates. As depicted in Fig. 3(b) (solid curves), we find that uniaxial [111] stress ($\alpha = 0$), although not achievable in the conventional DAC setup, yields the largest NV contrast and that there exists a strong correlation between the upper ISC rates (dashed curves) and the predicted contrast (solid curves).

To investigate these predictions, we directly measure [111]-NV contrast [extracted from Rabi oscillations, Fig. 3(d)] as a function of pressure in a (111)-cut diamond with NV centers implanted ~ 50 nm below the culet surface [Fig. 3(c)]. By carefully measuring the stress tensor at each pressure, we estimate the degree of hydrostaticity to be $\alpha \approx 0.73$. In addition, we also compare our predictions to two other sets of experimental NV-DAC measurements with differing degrees of hydrostaticity: (i) a nearly hydrostatic ($\alpha \approx 1$) measurement using NV's contained within a nanopillar fabricated at the center of (100)-cut DAC [16], and (ii) a measurement with $\alpha \approx 0.57$ [18] that also utilizes NVs in a (111)-cut DAC. As illustrated in Fig. 3(b), all three sets of experimental measurements are in semi-quantitative agreement with our *ab initio* predictions. Interestingly, since our calculations consider only a single NV center, this suggests that the contrast enhancement from utilizing a

(111)-cut anvil is *intrinsic* to the [111]-oriented NV itself, ruling out previous interpretations based on the “darkening” of non-[111] NVs [18].

Microscopic origin of stress-induced positive NV contrast—Let us now turn to a second puzzle regarding the NV's contrast in high-pressure experiments, namely, the observation of contrast inversion [depicted in Fig. 1(c) for a (100)-cut anvil] [63]. As previously discussed, conventional “negative” contrast occurs because the NV becomes optically polarized to the “bright” $|m_s = 0\rangle$ spin state, and the applied microwave transfers population into the comparatively “dark” $|m_s = +, -\rangle$ states. Conversely, “positive” contrast suggests that the NV is becoming polarized into a dark state. This hypothesis, as we will see, requires the lower ISC to develop a strong spin selectivity toward population transfer into the dark states. Crucially, this selectivity is made possible by symmetry-breaking stresses, which open new ISC transitions within the NV's optical cycle, e.g., $\Gamma_z^{\text{upper}}, \Gamma_z^{\text{lower}}$ [Fig. 2(b)].

To this end, let us begin by understanding the effect of symmetry-breaking stresses on the upper ISC rates. Under uniaxial [100] stress, the 3E sextuplet is split into two well-separated triplets, as the e orbital degeneracy is lifted [Fig. 2(b)]. Figure 4(a) depicts the upper ISC rates [defined in Fig. 2(b)] as a function of increasing [100] stress. The transition rate from $|m_s = +\rangle$ ($|m_s = -\rangle$) monotonically increases (decreases) with stress and is still primarily driven by the vibrational overlap between 3E_y (3E_x) and 1A_1 . This is because the 3E_x branch is rapidly detuned from 1A_1 (and vice versa for 3E_y). Meanwhile, the symmetry-breaking stress enables a nonzero Γ_z^{upper} , which connects $|m_s = 0\rangle$ to 1A_1 . Most importantly, we find that Γ_z^{upper} remains the smallest [Fig. 4(a)] throughout the entire pressure range investigated, confirming that the $|m_s = 0\rangle$ state remains the brightest.

Next, we turn to the fascinating case of the lower ISC. Much like 3E , the reduction of symmetry lifts the orbital degeneracy of 1E . Stress serves to progressively increase (decrease) the vibronic overlap between 1E_x (1E_y) and 3A_2 . In Fig. 4(b), we plot all three rates from the 1E singlet into the 3A_2 manifold. Among them, the behavior of Γ_z^{lower} is particularly intriguing owing to its non-monotonic behavior versus stress: It exhibits a significant dropoff beyond $\sigma_{ZZ} = 50$ GPa, and then increases sharply again after ~ 100 GPa. Somewhat remarkably, this results from the emergence of a new stress-induced spin-orbit channel from 1E_x , which *destructively interferes* with the existing Jahn-Teller-based channel (see Supplemental Material [31]). Above ~ 100 GPa, this new channel dominates, rapidly restoring the Γ_z^{lower} transition rate. Comparatively, Γ_z^{lower} progressively increases with stress and becomes dominant at $\sigma_{ZZ} \approx 60$ GPa, making the lower ISCs favor the dark spin state $|m_s = -\rangle$.

To this end, we observe a reversal of the spin selectivity of the lower ISC with symmetry breaking stress. By

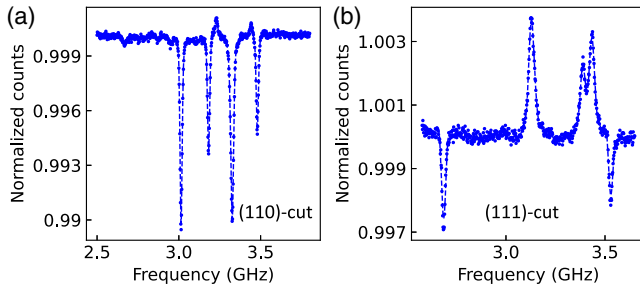


FIG. 5. Positive contrast observed from ODMR measurements performed on NV centers in (a) (110)-cut anvil at 300 K, 25 GPa, $B_z = 85$ G [12], and (b) (111)-cut anvil at 30 K, 28 GPa, $B_z = 150$ G. For the latter, the positive contrast originates from the non-[111] NV centers.

combining all of the aforementioned ISC rates, we compute the NV's spin polarization [Fig. 4(c)] and fluorescence contrast [Fig. 4(d)]. For small stresses, the NV remains initialized to the $|m_s = 0\rangle$ state, while fraction initialized into the $|m_s = -\rangle$ state begins to increase at $\sigma_{ZZ} \sim 25$ GPa. At even larger stresses, $\sigma_{ZZ} \gtrsim 65$ GPa, the NV becomes dominantly polarized to the $|m_s = -\rangle$ state. This represents the complex interplay between several symmetry-breaking-stress-induced modifications to both the upper and lower ISCs. From the perspective of contrast, once the NV becomes initialized into the dark $|m_s = -\rangle$ state, the resulting ODMR spectrum will naturally exhibit positive contrast [inset, Fig. 4(d)]. Interestingly, our theory predicts that while the ODMR peaks may exhibit contrast inversion, their positions still encode the same spectral content of the ground-state spin sublevel splittings.

Our theoretical framework also implies the existence of positive contrast in more general stress conditions. In particular, the interplay between symmetry breaking stresses and/or transverse magnetic fields can serve to hybridize excited state spin sublevels, enhancing the effective Γ_z^{upper} and thereby promoting ground state polarization inversion. To probe this generality, we perform high-pressure NV measurements in both a (110)-cut and a (111)-cut anvil. As shown in Fig. 5, in the presence of an external magnetic field, we observe contrast inversion for both settings. Transverse magnetic fields may promote contrast inversion by enhancing Γ_z^{upper} via excited state spin mixing; a detailed analysis of this process will be presented in a future work.

Conclusion and outlook—Our work opens the door to a number of intriguing future directions. First, our computational framework can readily be generalized to accommodate a wide array of environmental conditions, e.g., temperatures, electrical and magnetic fields, and stress environments. Moreover, our method can also be adapted to the vast emerging landscape of solid state spin defects [66–69], and will help identify candidates that are viable under extreme conditions. Second, while we have provided a general framework for the emergence of contrast

inversion, the details of how and when such inversion occurs in different anvil cuts remains an open challenge. Third, we motivate the deliberate engineering of stress environments to promote metrological sensitivity, i.e., $\alpha = 0$, which may be obtained with a uniaxial press [70]. Finally, the phenomenon of positive contrast suggests the use of symmetry-breaking stresses as another tuning parameter for defect physics. Modifying the polarization dynamics of the NV center and other spin defects may prove useful in the context of quantum sensing.

Note added—During the completion of this work, a related manuscript appeared [71], which explores the contrast inversion of the NV center under pressure in a (111)-cut anvil. The authors also conclude that this inversion arises from a reversal in spin polarization.

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Data availability—All data needed to reproduce the results in this work are available online [72].

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End Matter

The pursuit of accurate and efficient methods for solving the electronic structure problem has been the holy grail of computational chemistry and condensed matter physics. One central physical quantity investigated in this work is the spin orbit coupling (SOC), which is numerically obtained using the complete active space self-consistent field (CASSCF) method [73] applied to a single-NV cluster [74–77], as shown in the inset of Fig. 3(a). This model has been widely adopted in several recent studies [77–79] due to its affordable computational cost and availability of codes [80].

During this research, a subset of the authors developed an alternative description [81] of the defect's SOC based on a quantum defect embedding theory (QDET) [82,83] which partitions the problem into separate calculations of the defect center and of the host material. Since this approach uses periodically repeated cells, the surrounding solid-state environment is represented more accurately. QDET is based on a Green's function formalism, and has been applied to study several defects in semiconductors [84,85] for quantum technologies. In this End Matter, we shall make a comparison of these two computational approaches,

showcasing their respective pros and cons, to guide future computational or applied research.

In Table S6 in Supplemental Material [31], we compare the upper ISC rates computed from these two methods with experimental measurements in ambient conditions, showing that the QDET approach [81] gives a better agreement with experiment, while the CASSCF@cluster approach gives results roughly one order of magnitude smaller than what is observed. The inaccuracy of the CASSCF approach can be traced back to an underestimation of $\lambda_{\perp}$ (see Table S4). By comparing the weights of different configurations in the many-body wavefunctions from Refs. [78]

and [81], we expect that this underestimation arises from the relatively large weight ($\sim 20\%$) of the $e_x^2 e_y^2$ configuration in the 1A_1 wave function from CASSCF (see Sec. S6.2 in Supplemental Material [31]). However, in this work, the relevant quantities in our calculations are the strain/stress susceptibilities of SOC's, and CASSCF@cluster provides a consistent way to estimate them, as we reach a semi-quantitative agreement with experimental ODMR measurements.

Future work will employ the QDET approach to refine the estimation of susceptibilities.